

3. Data Types in GIS

Objectives

- . You should be able to choose organization structure for data, and create it in GIS
- . You should be able to choose right data types for data

Data & Meta Data

Data

Any collection of related facts arranged in a particular format; often, the basic elements of information that are produced, stored, or processed by a computer.

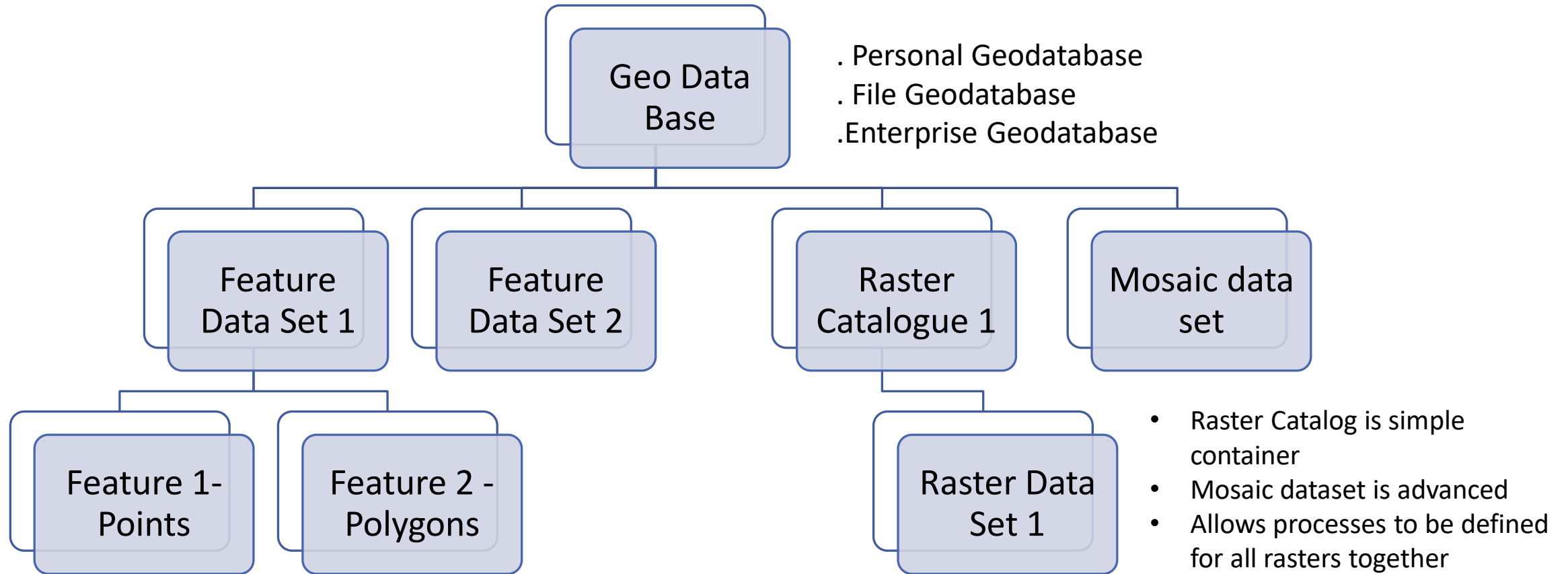
Metadata

Information that **describes the content, quality, condition, origin, and other characteristics of data** or other pieces of information.

Metadata for spatial data may describe and document its **subject matter; how, when, where, and by whom the data was collected**; availability and distribution information; its projection, scale, resolution, and accuracy; and its reliability with regard to some standard.

Metadata consists of properties and documentation. Properties are derived from the data source (for example, the coordinate system and projection of the data), while documentation is entered by a person (for example, keywords used to describe the data).

Structured Data Storage System

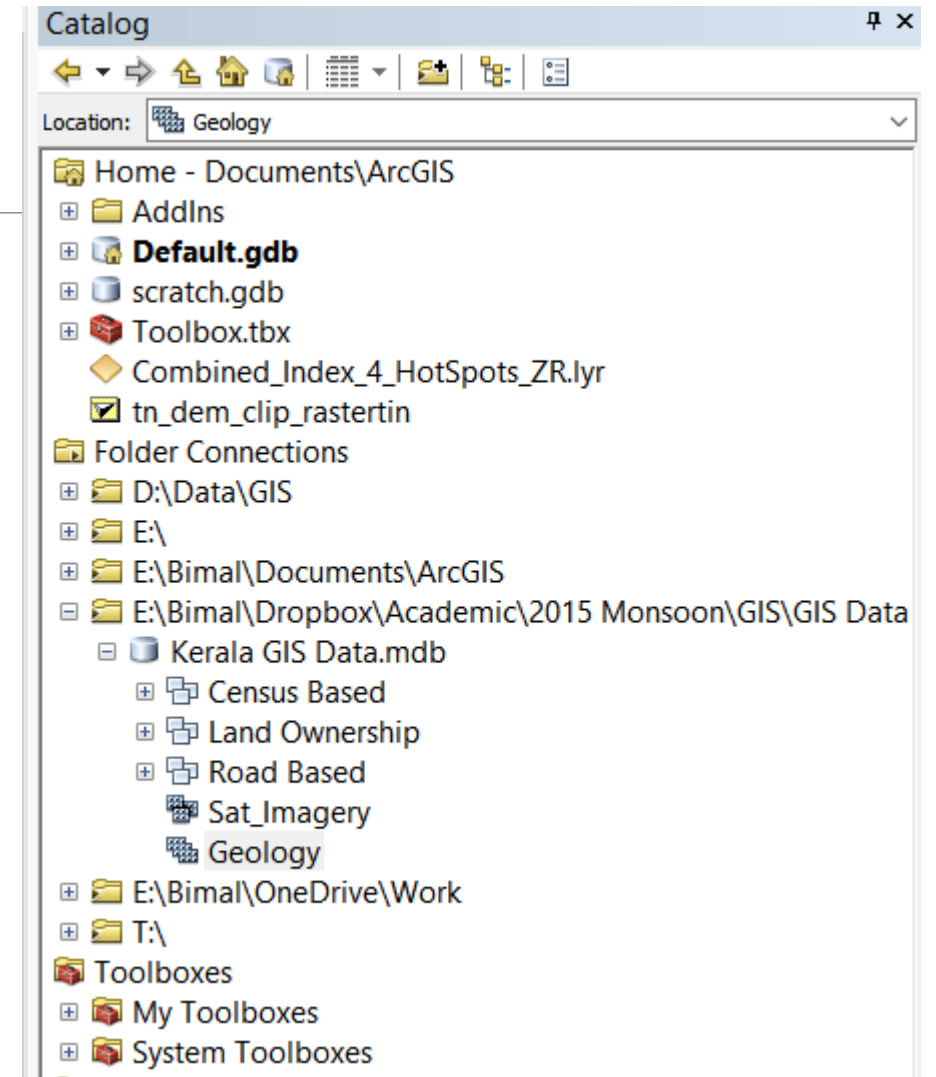


How to make them?

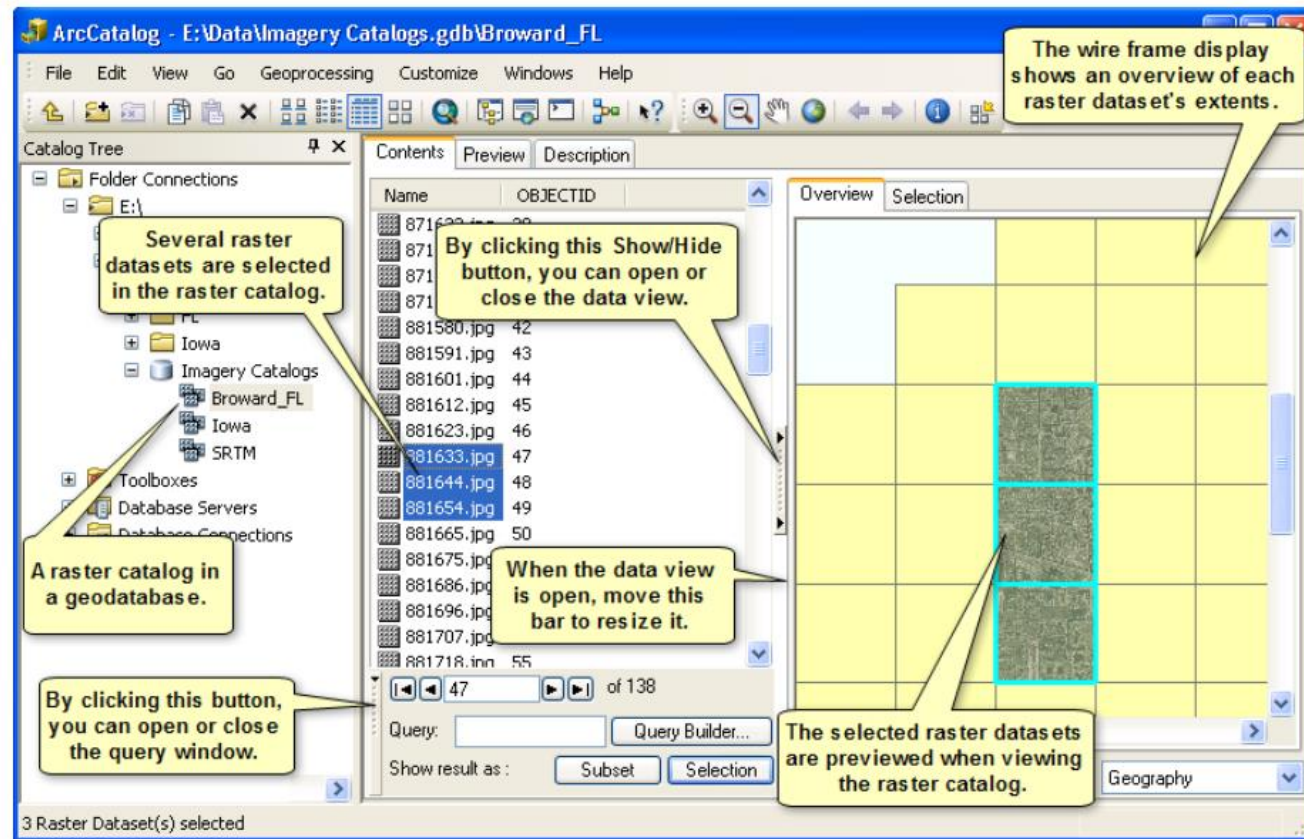
Use Arc Catalogue to Create Empty Data Structure

Add it to ArcMap

Insert data by digitising, analysis or other operations

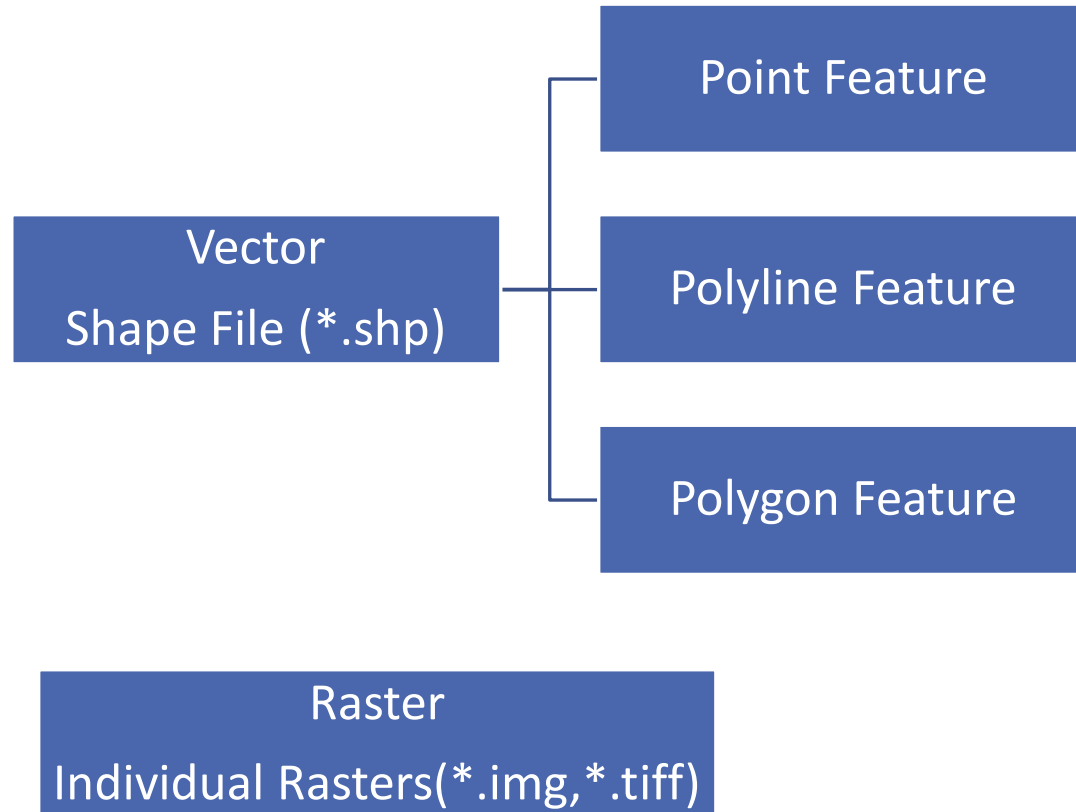


Raster Catalog



Exploring a raster catalog in ArcCatalog

Unstructured Data Storage

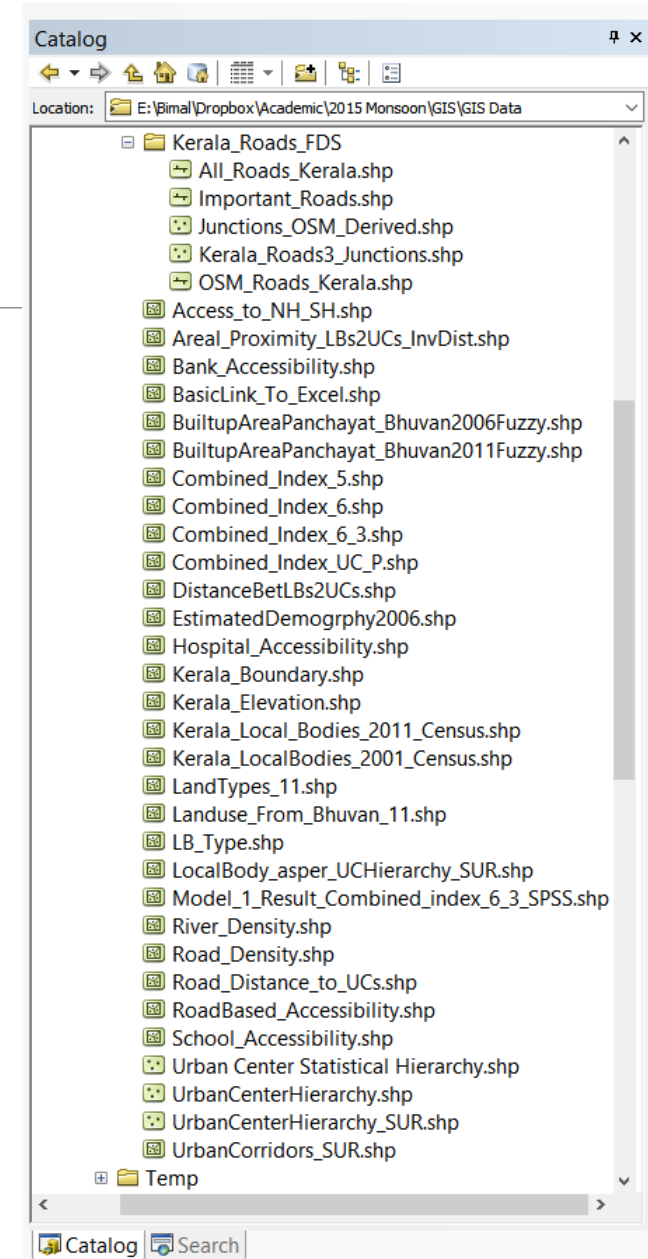


Advantages of Structured storage

- Easy to manage
- Essential for certain processes (Network analysis, Topology etc)
- OS treats them as single file

Advantages of Unstructured Storage

- OS treats them as multiple files – easy to manage individual features

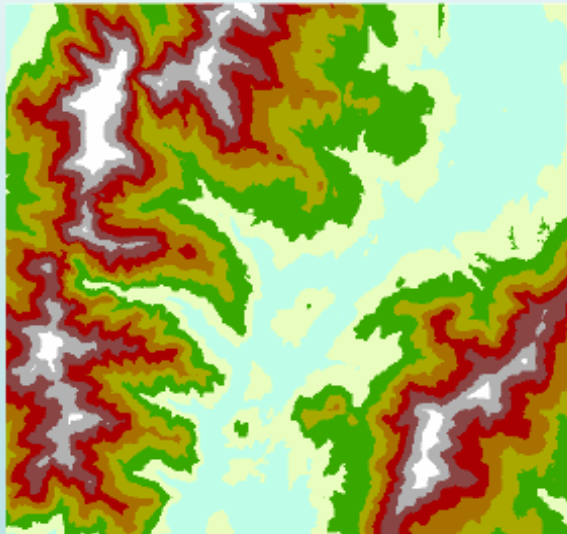




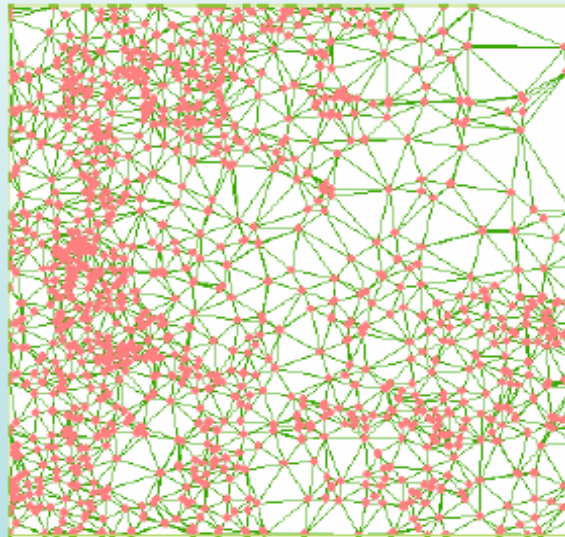
TIN – Triangulated Irregular Network

- Great for representing topographic surfaces
- Surface is represented as contiguous, non-overlapping triangles (facets)
- Better than raster grids for “rough” terrain with cliffs and streams
- Unlike systematic DEM grids, these are irregularly distributed

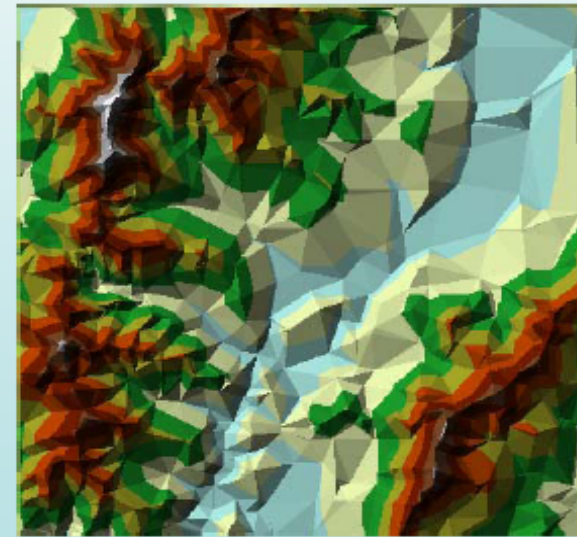
(DEM -- Digital Elevation Model)



DEM raster



TIN triangles



TIN surface

Attribute Data

- Short integer
- Long Integer
- Float
- Double
- Text
- Date

Name	Specific range, length, or format	Size (Bytes)	Applications
Short integer	-32,768 to 32,767	2	Numeric values without fractional values within specific range; coded values
Long integer	-2,147,483,648 to 2,147,483,647	4	Numeric values without fractional values within specific range
Single precision floating point number (Float)	Approximately -3.4E38 to 1.2E38	4	Numeric values with fractional values within specific range
Double precision floating point number (Double)	Approximately -2.2E308 to 1.8E308	8	Numeric values with fractional values within specific range
Text	Up to 64,000 characters	Varies	Names or other textual qualities
Date	mm/dd/yyyy hh:mm:ss A/PM	8	Date and/or time

Table

Export_Output_10

Add Field

Name: Field1

Type: Short Integer

Field Properties

Precision

OK Cancel

Export_Output_10

(0 out of 1051 Selected)

UC1_ID	UC1_POPDen	UC1_BUDens	A_DistUC1	UC2_ID	UC2_PopDen	UC2_BUDens	A_DistUC2	RUC1_ID	RUC1_PDens	RUC1_BuDen	RUC1_TT
107	4195.65	0.791145	103701	1	3259.08	0.141448	24782.3	19	3259.08	0.141448	33.9785
107	4195.65	0.791145	103647	2	3259.08	0.141448	22885.6	19	3259.08	0.141448	27.033
107	4195.65					0.141448	19147.2	19	3259.08	0.141448	29.3811
107	4195.65					0.141448	14505.3	0	0	0	0
107	4195.65					0.141448	12931.2	19	3259.08	0.141448	20.9307
107	4195.65					0.141448	14391.7	19	3259.08	0.141448	30.644
107	4195.65					0.141448	6444.6	19	3259.08	0.141448	25.6731
107	4195.65					0.141448	8528.09	19	3259.08	0.141448	16.2976
107	4195.65					0.141448	19153.1	19	3259.08	0.141448	72.0197
107	4195.65					0.141448	7202.59	19	3259.08	0.141448	19.3929
107	4195.65					0.141448	19619.5	19	3259.08	0.141448	47.4472
107	4195.65					0.141448	13232.2	19	3259.08	0.141448	31.0105
107	4195.65					0.141448	14412.9	19	3259.08	0.141448	34.3212
107	4195.65					0.141448	538.288	19	3259.08	0.141448	8.27969
107	4195.65					0.141448	1290.94	19	3259.08	0.141448	7.88285
107	4195.65					0.141448	3234.01	19	3259.08	0.141448	6.08355
107	4195.65					0.141448	8133.76	19	3259.08	0.141448	22.7935
107	4195.65					0.141448	20670.4	19	3259.08	0.141448	40.2627
107	4195.65					0.141448	100	19	3259.08	0.141448	2.00372
107	4195.65					0.141448	9906.14	34	3133.29	0.058186	31.132
107	4195.65					0.141448	955.505	19	3259.08	0.141448	12.6519
107	4195.65					0.058186	26131.2	19	3259.08	0.141448	55.8689
107	4195.65	0.791145	68898.9	23	3259.08	0.141448	4946.34	19	3259.08	0.141448	16.3897
107	4195.65	0.791145	58746.3	24	3133.29	0.058186	17837.4	19	3259.08	0.141448	50.7098



Q&A

1. Suggest a Personal GeoDataBase structure for your RWA Study
2. What data types will you select for the following attribute data?
 1. House numbers
 2. House owners name
 3. Family income
 4. Floor area